

# The nim-factorial of $2^n - 1$ is 1: a proof of Layman's conjecture on OEIS A059970

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## Abstract

Let  $a(n) = 1 \otimes 2 \otimes \cdots \otimes n$  denote the nim-factorial, where  $\otimes$  is Conway's nim-multiplication (OEIS A059970). In March 2001 J. W. Layman conjectured that  $a(2^n - 1) = 1$  for all  $n$ , reporting verification for  $n \leq 16$ . We prove the following stronger and cleaner statement: for every  $n \geq 0$ , the nim-product of all nonzero nimbers below  $2^n$  equals 1. The conjecture is the special case, since  $\{1, \dots, 2^n - 1\}$  is exactly that set. The proof is a two-line induction once one observes that  $\{0, 1, \dots, 2^n - 1\}$  splits as a union of additive cosets of the Conway field of order  $2^{2^k}$ , on each of which the product collapses by Wilson's theorem for finite fields.

## 1 The sequence and the conjecture

Conway's nim-addition  $\oplus$  on the non-negative integers is bitwise exclusive-or, and nim-multiplication  $\otimes$  is defined by the mex rule  $a \otimes b = \text{mex}\{a' \otimes b \oplus a \otimes b' \oplus a' \otimes b' : a' < a, b' < b\}$  [1, Ch. 6]. OEIS sequence A059970 is the nim-factorial

$$a(1) = 1, \quad a(n) = n \otimes a(n-1) \quad (n > 1),$$

contributed by John W. Layman on 5 March 2001, with first terms 1, 2, 1, 4, 2, 11, 1, 8, 5, 9, 2, 4, 9, 4, 1, 16,  $\dots$

The entry carries the comment

**Conjectures:** (1) Nim-Factorial( $2^n - 1$ ) = 1 (verified for  $n = 1, 2, 3, \dots, 16$ ). (2) Nim-Factorial( $2^n + 2^{n-1} - 1$ ) = 2 (verified for  $n = 1, 2, 3, \dots, 15$ ).

This note settles part (1). As of the “Last modified” line on the live entry (23 August 2026) both statements are still recorded as unproven conjectures, no proof appears in the entry's links or programs, and a search of the literature on nimber arithmetic turns up nothing on products of consecutive nimbers.

## 2 Two standard facts, and two lemmas

We use three facts from Conway [1, Ch. 6]. Write  $F_k = 2^{2^k}$  for the  $k$ -th Fermat 2-power.

(N1) The set  $K_k = \{0, 1, \dots, F_k - 1\}$  of nimbers below  $F_k$  is a field of order  $F_k$  under  $(\oplus, \otimes)$ ; that is,  $K_k \cong \text{GF}(2^{2^k})$ , and  $K_0 \subset K_1 \subset K_2 \subset \dots$ .

(N2)  $F_k \otimes a = F_k \cdot a$  (ordinary product) for every  $a < F_k$ .

(N3)  $F_k \otimes F_k = \frac{3}{2}F_k = F_k \oplus \frac{1}{2}F_k$ .

Throughout, fix  $k$  and abbreviate  $F = F_k$ ,  $K = K_k$ ,  $q = |K| = F$ , and let  $\mathbb{F} = K_{k+1} \cong \text{GF}(q^2)$ , so that  $K \subset \mathbb{F}$  and  $F \in \mathbb{F} \setminus K$ . All powers below are nim-powers taken in  $\mathbb{F}$ .

**Lemma 1.**  $F^q = F \oplus 1$ .

*Proof.* By (N3),  $F \otimes F = F \oplus c$  with  $c = \frac{1}{2}F \in K$ . Hence  $F$  is a root of  $X^2 \oplus X \oplus c \in K[X]$ . Since  $F \notin K$  this quadratic is irreducible over  $K$ , so its two roots lie in  $\mathbb{F} = \text{GF}(q^2)$  and are interchanged by the nontrivial element  $x \mapsto x^q$  of  $\text{Gal}(\mathbb{F}/K)$ . The two roots sum to the coefficient of  $X$ , namely 1, so they are  $F$  and  $F \oplus 1$ . Therefore  $F^q = F \oplus 1$ .  $\square$

**Lemma 2.** For every  $z \in \mathbb{F}$ ,  $\bigotimes_{b \in K} (z \oplus b) = z^q \oplus z$ . Consequently, for every  $c \in K$ ,

$$\bigotimes_{b \in K} ((c \otimes F) \oplus b) = c.$$

*Proof.* In  $K[X]$  we have  $\prod_{b \in K} (X - b) = X^q - X$ ; evaluating at  $z$  and using characteristic 2 gives the first identity. For the second, put  $z = c \otimes F$ . Since  $c \in K$  we have  $c^q = c$ , so by Lemma 1

$$z^q = c^q \otimes F^q = c \otimes (F \oplus 1) = (c \otimes F) \oplus c = z \oplus c,$$

whence  $z^q \oplus z = c$ .  $\square$

### 3 The theorem

**Theorem 3.** For every integer  $n \geq 0$ ,

$$\bigotimes_{v=1}^{2^n-1} v = 1.$$

*Proof.* Induction on  $n$ . For  $n = 0$  the product is empty and equals 1.

Let  $n \geq 1$  and let  $2^k$  be the largest power of 2 with  $2^k \leq n$ ; put  $m = n - 2^k$ , so  $0 \leq m < 2^k$ . Keep the notation  $F = F_k = 2^{2^k}$ ,  $K = K_k$ ,  $q = |K| = F$  of the previous section. Note  $2^m \leq F$  because  $m < 2^k \leq 2^{2^k}$ , and  $2^n = 2^{2^k+m} = 2^m \cdot F$ .

Every integer  $v$  with  $0 \leq v < 2^n$  has a unique representation  $v = a \cdot F + b$  with  $0 \leq a < 2^m$  and  $0 \leq b < F$ . Because  $b < F$  the ordinary sum  $a \cdot F + b$  has no carries, so it equals  $a \cdot F \oplus b$ ; and  $a < 2^m \leq F$ , so  $a \cdot F = a \otimes F$  by (N2). Hence

$$\{0, 1, \dots, 2^n - 1\} = \{ (a \otimes F) \oplus b : 0 \leq a < 2^m, b \in K \},$$

and the representation is unique. Splitting the product according to  $a$ ,

$$\bigotimes_{v=1}^{2^n-1} v = \left( \bigotimes_{\substack{b \in K \\ b \neq 0}} b \right) \otimes \left( \bigotimes_{a=1}^{2^m-1} \bigotimes_{b \in K} ((a \otimes F) \oplus b) \right).$$

The first bracket is the product of all nonzero elements of the finite field  $K$ , which is  $-1 = 1$  in characteristic 2 (Wilson's theorem for finite fields). By the second part of Lemma 2 the inner product in the second bracket equals  $a$ . Therefore

$$\bigotimes_{v=1}^{2^n-1} v = \bigotimes_{a=1}^{2^m-1} a,$$

which equals 1 by the induction hypothesis, since  $m < 2^k \leq n$ .  $\square$

**Corollary 4.**  $a(2^n - 1) = 1$  for every  $n \geq 1$ , where  $a$  is A059970. This is Layman's conjecture (1).

*Proof.*  $a(2^n - 1) = \bigotimes_{v=1}^{2^n-1} v$ , which is 1 by Theorem 3. □

**Remark.** Theorem 3 says more than the conjecture only in the sense that it identifies the mechanism:  $\{1, \dots, 2^n - 1\}$  is the set of nonzero elements of an  $\mathbb{F}_2$ -subspace of a Conway field, and the leading-bit recursion  $\bigotimes_{v < 2^n} v = \bigotimes_{v < 2^m} v$  with  $m = n - 2^{\lfloor \log_2 n \rfloor}$  strips the binary expansion of  $n$  one bit at a time. The statement is elementary: it uses nothing beyond Conway's construction of the nimber field and Wilson's theorem. The appropriate destination is an OEIS comment, not a journal.

## 4 Verification

A verification script, written separately from this note, was used to check every assertion made above. It (i) implements nim-multiplication twice by genuinely different algorithms — Conway's recursion on Fermat 2-powers, and a direct tabulation from the mex definition — and confirms that they agree on  $\{0, \dots, 31\}^2$  with 0 mismatches; (ii) reproduces the first 70 published terms of A059970 with 0 mismatches; (iii) confirms  $F \otimes F = F \oplus \frac{1}{2}F$  for  $k = 0, \dots, 4$ ,  $F \otimes a = F \cdot a$  for  $a < F$  and  $k = 0, \dots, 3$ , and  $F^q = F \oplus 1$  for  $k = 0, \dots, 3$ ; (iv) confirms Lemma 2 in 148 instances of the first identity and all 22 instances of the second with  $k \leq 2$ ; (v) confirms Theorem 3 directly for  $n = 0, \dots, 14$ ; and (vi) computes  $a(2^n - 1)$  from the defining recursion  $a(n) = n \otimes a(n - 1)$  for  $n = 1, \dots, 18$ , obtaining 1 in every case. The largest index reached was  $a(262143) = 1$ ; the conjecture as posted was verified only to  $n = 16$ .

## References

- [1] J. H. Conway, *On Numbers and Games*, 2nd ed., A K Peters, 2001, Chapter 6.
- [2] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.
- [3] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A059970>, entry A059970, consulted 23 August 2026.